

From Code to the Real World

ESP8266 • Scratch • Wokwi • APIs

CHITCHAT / DEMO DAY

No experience required.

Today's promise

You will see software move through three stages:

IDEA → LOGIC → VIRTUAL HARDWARE → REAL-WORLD ACTION

We will build one simple project together, then turn the same idea into two event machines.

Three projects / one journey

- | | | |
|---|--------------|--------------------------------|
| 1 | TOUCH LED | audience builds the logic |
| 2 | WEATHER BOX | rain event → umbrella reaction |
| 3 | FOOTBALL BOX | goal event → celebration |

One hands-on activity. Two fast visual WOW demos.

Start with Scratch

Before wiring anything, describe the behaviour:

```
WHEN touched
  SAY "I felt that!"
  TURN LED ON
ELSE
  TURN LED OFF
```

Scratch makes the decision visible. The board comes next.

Project 1 / Touch LED

TOUCH SENSOR → ESP8266 → ONBOARD LED
input decision output

Ask the room:

“What should happen when I touch the sensor?”

No cloud. No AI. Just **input** → **process** → **output**.

Audience build

```
IF touched
  LED ON
ELSE
  LED OFF
```

Then change one thing:

- touch once → toggle
- blink three times
- add a sound

The audience becomes the developer.

Same idea / new tool

SCRATCH BLOCKS



ARDUINO C/C++



ESP8266 PIN

Show only the few lines that read the input and control the output. Explain that **MicroPython** is a friendly alternative for a later workshop.

LIVE NOW / Project 2

Weather Event Box 

Rain becomes physical

PHONE BUTTON or WEATHER API



VIRTUAL ESP32



BLUE LIGHT BUZZER FLAG

Press **Rain expected** in the browser simulator.

Global data → local action.

LIVE NOW / Project 3

Football Event Box 

A goal happens somewhere else

PHONE BUTTON or FOOTBALL API



GOAL!



VIRTUAL ESP32



FLASH



SOUND

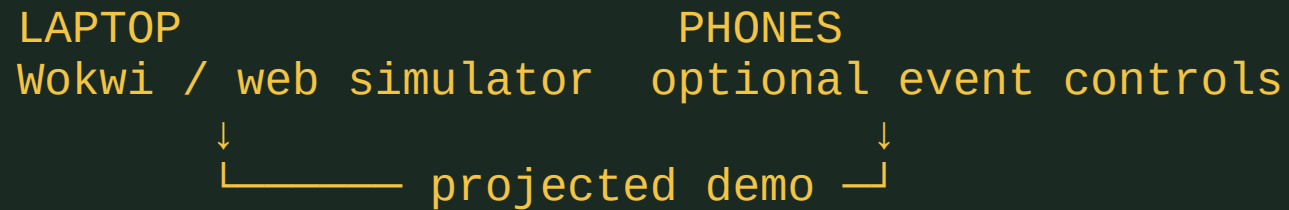


FLAG

Press **Football goal**.

The box celebrates here.

How the simulator is set up



The laptop buttons are the reliable controller. Phone pages are optional and currently run independently; shared phone control requires a small server bridge.

The reusable pattern

READ → INTERPRET → DECIDE → ACT

The input can be:

- a touch sensor
- a weather API
- a football event
- a local soil sensor
- an AI recommendation

The pattern stays the same.

What comes next?

SCRATCH → WOKWI → ARDUINO → MICROPYTHON

↓
APIs • MQTT • AI

The bootcamp turns today's three small ideas into projects you can build yourself.

Your turn

What would you build if your code could control the real world?

VibeCode.Tours × Open Hardware